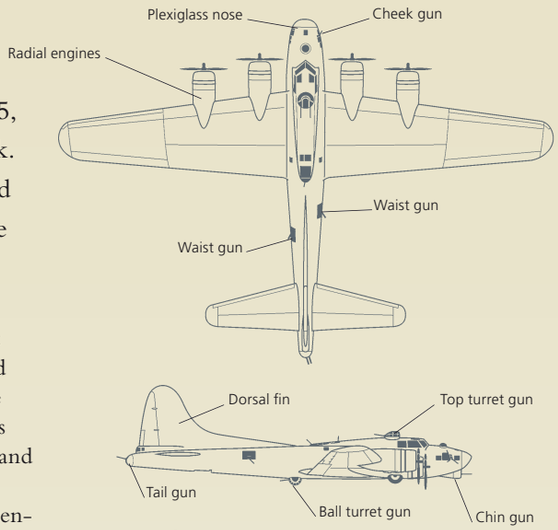
B-17 BOMBER

When the Boeing B-17 bomber first flew as the Model 299 prototype on 28 July 1935, it was immediately dubbed "the Flying Fortress" by journalists, a nickname that stuck. The B-17 belonged to a new generation of all-metal monoplane aircraft with enclosed cockpits. Large and fast by the standards of its time, the four-engined bomber became a mainstay of America's strategic bombing force of World War II.

The B-17 went through radical modifications between the prototype stage and the entry of the US into World War II. The first mass-produced version, the B-17E, emerged in September 1941, and B-17Fs and B-17Gs were introduced in 1942. Around two-thirds of all B-17s built were the G version. The B-17G had up to 13 machine-guns (hence its nickname), and its maximum speed was 462kph (287mph) at 7,600m (25,000ft), although the normal operational speed flying in mass formation was a more modest 290kph (180mph).

On a short-range flight the aircraft could lift a 7,983kg (17,600lb) bomb load, but the need to carry large amounts of fuel for long-range sorties meant that in practice the payload was typically between 1,814 and 2,724kg (4,000 and 6,000lb). The B-17 was never a comfortable aircraft to fly in, but it was popular with its ten-man crews because of its legendary ability to survive heavy punishment. Nonetheless, losses were severe; around 4,750 B-17s were lost on combat missions in the course of the war, more than a third of the total of 12,761 built.



Bomber profile
The B-17 had a wingspan of 31.62m (103ft 9in) and measured 22.78m (74ft 9in) from nose to tail. The massive dorsal fin gives it its distinctive profile.



> "WE LIVED, SLEPT, ATE, WORKED, AND PLAYED TOGETHER. WE WOULD SHARE OUR LIVES UNTIL DEATH OR THE WAR'S END."
>
> **LIEUTENANT ROLAND PEPIN**, B-17 NAVIGATOR

Flying boots
Aircrew wore sheepskin-lined rubber-soled boots to keep their feet warm at high altitude. They also had electrically-heated flying suits that they could plug in on board the aircraft.

Lifejacket
A "Mae West" inflatable jacket was among the array of life-saving equipment provided for aircrew.



Aircraft nose
The Plexiglass nose of the B-17 provided a view for the bombardier, who guided the aircraft to its target.



Chin gun
This is one of the remote-controlled chin guns operated by the bombardier. They were introduced to counter head-on attacks.



Radial engine
Each of the B-17's four engines generated 1,200hp. They were fitted with turbo-superchargers to improve performance at high altitude.



Wing root
The B-17 was mass produced and so construction methods were kept as simple as possible. Riveted aluminium sheets were a major component.